THE UNIVERSITY OF HONG KONG

FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE

COMPP7103 Data Mining
(Subclass B)

Date: 10 May, 2023 **Time: 6.30pm – 8.30pm**

University No: | | | | | | | | | | |

This is a closed book examination.

There are 4 questions in total. The first question gives 10/100 points, the second one 20/100, the third one 30/100, and the fourth one 40/100.

Answer ALL 4 questions.

Write your answers on the exam paper in the corresponding space after the question.

Write your university number on every page.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Brand and Type of calculator: ____________________

University No: ____________________

Question 1 (General questions, 10/100).

Provide a short answer to each of the following questions (1-2 sentences):

a) (2/100) Explain the main idea behind random forests and why they work well in practice.

b) (2/100) Why is it preferable that a decision tree be not too large?

c) (3/100) Which independence assumption we use in the Naive Bayes classifier?

d) (3/100) Consider the variant of PageRank with restart (aka personalized PageRank), where only one seed node u is given. Is the node closest to u in terms of shortest path distance the one with the largest PR score?

Question 2 (PageRank, 20/100)

Consider the directed graph G depicted below. Without introducing random jumps:

a) (10/100) compute the PageRank vector of G, that is, the vector that the PageRank algorithm converges to when G is provided in input. Report the PageRank vector below

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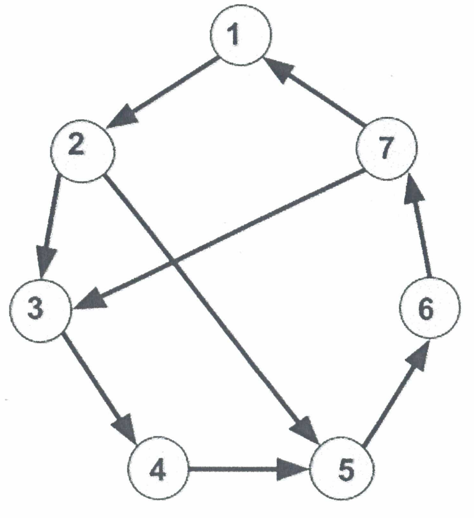
b) (10/100) Report the first two vectors obtained as a result of the PageRank algorithm with input G starting from the vector (1/7,1/7,1/7,1/7,1/7,1/7,1/7).



Question 3 (decision tree, 30/100).

Consider the following data about 9 samples of mushrooms with features: spotty, smelly, smooth, and edible (the class).

Sample	spotty	smelly	smooth	**edible**
A	0	0	1	1
B	0	1	1	1
C	0	1	1	1
D	1	0	0	0
E	0	1	0	0
F	1	1	1	1
G	1	0	1	0
H	1	1	1	0
I	1	1	0	1

a) (10/100) construct a decision tree using the following rules: (i) at each step compute the gini index for every possible split considering all attributes (except "Sample") and select the split with the best gini value; (ii) stopping rule: when the gini value of a node is zero or no further split is possible; (iii) the class of a leaf node is determined by the majority rule (breaking ties arbitrarily). Draw the decision tree so constructed below:

b) (10/100) Report the class value of ("L", 0,1,0).

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c) (10/100) Prune the tree so as to improve its generalization error = 0.5 * number of leaves + training error. Draw the final decision tree below (if different from a)).

Question 4 (k-means, 40/100) Using the same terminology adopted in our course, we shall refer to the ``k-means'' algorithm as the algorithm that initializes the centroids randomly, followed by a ``refinement phase'' where the clusters are improved further. We shall refer to ``k-means++'' as the algorithm that only selects the centroids, so as to provide some theoretical guarantees. Consider the following points in the 2D Euclidean space: p1=(0,0), p2=(0,1), p3=(0,2), p4=(2,0), p5=(3,0), p6=(4,1), p7=(5,0), p8=(7,0), p9=(8,0), p10=(8,1). Let k=3.

a) (10/100) Run the k-means++ algorithm to select the initial centroids, assuming that 1) p4 is selected as first centroid 2) the remaining two centroids are chosen assuming that at each step the point with 3rd largest probability is selected by k-means++ (breaking ties arbitrarily). In other words, let q1,q2,q3, .. ,qn be the input points sorted non-increasingly according to their probability of being selected at a given step of k-means++. Then, the point q3 is going to be selected at that step. Which centroids have been selected at the end of this initialization step?

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b) (10/100) What is the probability that p5 is selected as centroid at step 3 of k-means++?

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c) (10/100) Run the refinement phase of the k-means algorithm until it terminates while using the centroids selected in a). What are the final clusters?

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d) (10/100) Consider the variant of the k-means algorithm, where 1) any given point p can be moved from cluster C with centroid c to a cluster C' with centroid c' even if d(p,c)=d(p,c'), 2) we stop as soon as we obtain the same *clustering* in two consecutive iterations. Show that this variant of k-means might never terminate by providing an example with at most 5 points in the 1-dimensional Euclidean space:

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